

CURE: Constrained Uncertainty Routing with Complementary Experts for Retrieved-Evidence Fact Verification

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Abstract

Fact verifiers often conflate two distinct decisions: which verdict is most likely and whether the retrieved evidence warrants a decisive verdict. We study this distinction in a fixed-corpus, text-evidence MOCHEG protocol. CURE (Constrained Uncertainty Routing with Complementary Experts) combines a verdict-trained direct expert with a rationale-trained complementary expert. Its *Asymmetric NEI Deferral* rule preserves the direct verdict by default and permits only a high-confidence complementary NEI prediction to defer the output. On frozen validation, this constrained rule improves Macro-F1 from 0.6920 to 0.7112 (+1.92 points; paired-bootstrap $P(\Delta > 0) = 0.9975$), whereas equal averaging and symmetric confidence routing do not. With frozen $K = 5$ and $\tau = .49$, the raw/strict P1 test point estimates are $+0.00977/+0.00943$ MF1 (95% CIs $[-.00128, .02109]/[-.00187, .02065]$; $P(\Delta > 0) = .9529/.9477$). These are directional rather than statistically resolved test results. A matched teacher-capacity ablation finds higher 3B-than-7B rationale-teacher point estimates, but its intervals also include zero, ruling out a simple teacher-scaling explanation rather than establishing a preferred capacity. Post-hoc diagnostic groups suggest a benefit for some NEI cases and identify retrieval misses as a failure mode. A separate validation-selected heterogeneous ensemble reaches 0.5551 Macro-F1 on raw P1, and a disagreement-distilled student offers a lower-latency extension.

1 Introduction

Automated fact checking is commonly decomposed into evidence retrieval and claim verification (Guo et al., 2022; Thorne et al., 2018). Modern systems increasingly retrieve heterogeneous evidence and use large language models, but two assumptions remain fragile: that all retrieved passages should be consumed by the verifier, and that

a model trained to predict a verdict is also a reliable detector of *retrieved-evidence sufficiency*. We focus on the latter distinction in a controlled text-evidence setting.

MOCHEG (Yao et al., 2023) provides a useful setting because it couples three-way truthfulness labels with textual, visual, and social evidence. We deliberately isolate its text-evidence component: all experts receive the same fixed-corpus retrieved passages, so that effects of supervision and decision policy are not confounded by live-web retrieval or visual modeling. This is therefore a fact-verification study on a multimodal benchmark, rather than a claim that the proposed system itself is multimodal.

Our central observation is that two verifiers have complementary error profiles. A direct verifier is strong when evidence appears decisive but can overcommit under non-decisive retrieved context. A rationale-trained expert is more likely to predict NEI, but is not uniformly more accurate and is not assumed to be a calibrated sufficiency estimator. We turn this complementarity into a constrained policy rather than averaging the experts. The direct expert remains the default; only a high-confidence complementary NEI signal can defer the decision. We call this operator *Asymmetric NEI Deferral* (AND). It is a class-constrained cross-expert decision rule, not Bayesian epistemic uncertainty estimation.

Our contributions are:

- CURE, a dual-expert formulation that combines direct verdict prediction with a rationale-trained complementary NEI signal;
- AND, a class-constrained cross-expert rule that permits only NEI deferral, and controlled ablations against rationale-trained-only, posterior averaging, and symmetric routing alternatives;

- post-hoc diagnostic analyses that identify NEI improvements and retrieval-miss failure cases, without using diagnostic labels as routing inputs; and
- a disagreement-distilled student as a secondary, lower-latency deployment extension.

We distinguish frozen-test evaluation from development diagnostics. The heterogeneous ensemble and AND answer different questions: the former tests broad probability complementarity, while the latter tests a fixed, class-constrained decision policy. Both are evaluated on their respective frozen test protocols without test-time parameter selection.

2 Related Work

Fact verification and evidence sufficiency. FEVER established evidence-supported three-way verification at scale (Thorne et al., 2018); LIAR studied short political claims (Wang, 2017); and MultiFC broadened evaluation across sources and domains (Augenstein et al., 2019). SciFact further couples scientific claim labels with evidence rationales (Wadden et al., 2020). These benchmarks make clear that label prediction, evidence retrieval, and the availability of a convincing rationale are not identical targets. Surveys consequently treat retrieval, evidence selection, reasoning, and calibration as coupled but distinct stages (Guo et al., 2022). UKP-Athene illustrates the role of multi-sentence textual entailment in evidence-based verification (Hanselowski et al., 2018).

Retrieval, selection, and reasoning. Evidence retrieval can dominate downstream verification (Zheng et al., 2024); evidence selection itself materially changes end-to-end performance (Samarinas et al., 2021). RAV combines retrieval with verification, while RAFTS synthesizes contrastive arguments from retrieved evidence (Yue et al., 2024). Complex-claim verification decomposes claims before retrieving web evidence (Chen et al., 2024a). These approaches study different evidence regimes from our P1 protocol: they motivate the separation between fixed-corpus evidence selection and live-web retrieval rather than serving as directly interchangeable leaderboard rows.

Rationales, selective prediction, and expert routing. Extractive and generated rationales have been studied as both explanations and learning signals (Lei et al., 2016; Atanasova et al., 2020);

ERASER emphasizes that plausibility and faithfulness require separate evaluation (DeYoung et al., 2020). Our matched-recipe control follows this caution: rationale supervision is not assumed to cause a verdict gain. AND is also related to selective prediction with a reject option (Geifman and El-Yaniv, 2019), but it does not learn a reject head or optimize risk-coverage. Instead, it compares a frozen direct posterior with a complementary expert and permits one class-specific deferral. This differs from sparsely gated mixture-of-experts routing (Shazeer et al., 2017): CURE evaluates all members sequentially and uses a fixed, interpretable decision rule rather than learned conditional computation. Because softmax confidence need not be calibrated (Guo et al., 2017), we report calibration diagnostics and explicitly compare against direct self-deferral.

MOCHEG and multimodal fact checking. MOCHEG introduced end-to-end multimodal fact checking and explanation generation (Yao et al., 2023). Recent systems pursue different ways to aggregate textual and visual evidence: LVL4M4FV uses large vision-language models with reranked evidence (Tahmasebi et al., 2024); MetaSumPerceiver creates claim-specific multimodal summaries (Chen et al., 2024b); HGTMF models high-order interactions with hypergraphs (Pang et al., 2025); MEVER jointly addresses retrieval, verification, and explanations (Zhang et al., 2026); and M-RAV augments cross-modal evidence alignment for zero-shot verification (Luu et al., 2027). AMuFC adaptively determines whether visual evidence is needed (Jung et al., 2026). These systems motivate the benchmark comparison, but our P1 method consumes retrieved text only. CURE consequently studies constrained routing between differently trained text verifiers, not multimodal fusion.

3 Task Definition and Protocols

Given a claim c and retrieved evidence $E = \{e_j\}_{j=1}^K$, predict $y \in \{\text{SUPPORTED}, \text{REFUTED}, \text{NEI}\}$. The principal P1 protocol uses MOCHEG’s fixed corpus and upstream text retrieval; it is not an open-web evaluation. The raw official test contains 2,442 claims. A strict robustness track removes eight verbatim cross-split duplicates and contains 2,434 claims. We report both tracks separately. Development validation contains 1,456 claims.

All routing thresholds are selected on validation

or train-only out-of-fold data. Test labels are never used for selecting a seed, threshold, or policy in frozen-test evaluation. Open-web experiments are not mixed into P1 tables.

4 Method

4.1 Direct and rationale-trained experts

The direct expert f_D receives a claim c and the top- K reranked passages E_K and predicts a three-way distribution $p_D(y \mid c, E_K)$. It is optimized only with verdict cross-entropy. The rationale-trained expert f_G has the same Qwen3-4B-LoRA backbone and the same inference prompt, but is additionally trained to generate a structured teacher rationale. Thus, it remains a verdict classifier at inference rather than a rationale generator. Its auxiliary targets are schema-validated against retrieved passages; this provenance check does not assert that $p_G(\text{NEI})$ is a calibrated measure of evidence sufficiency.

The primary rationale targets are generated once for training claims by Qwen2.5-7B-Instruct using the *gold training verdict* and the system-retrieved top-five passages. The teacher is instructed to flag a rationale when those passages cannot justify that training verdict. A valid target contains a retrieval-validation flag, verdict, key-passage identifiers, free-text reason, and missing-information field. Generation is deterministic (greedy decoding, maximum 512 new tokens), and schema validation rejects malformed key-passage references and unsupported structured outputs. Explanation loss is applied only to valid, retrieval-supported targets. A later matched capacity ablation replaces only this teacher with Qwen2.5-3B-Instruct. This use of training labels is ordinary supervised learning; no validation or test label, and no gold evidence passage, is included in a validation/test prompt.

For a training example i , let z_i be the verdict target and r_i the structured rationale target. The two expert objectives are

$$\begin{aligned}\mathcal{L}_D &= - \sum_i \log p_D(z_i \mid c_i, E_i), \\ \mathcal{L}_G &= \mathcal{L}_D + \lambda_r \mathcal{L}_{\text{rationale}}(r_i), \quad \lambda_r = 0.25.\end{aligned}\tag{1}$$

The rationale term is token-level cross-entropy over the validated structured target. Both experts begin from the same base model. We report two recipe controls: a five-fold train-only OOF explanation-supervision contrast under a common recipe, and

a fresh direct-only control matching the headline rationale-trained data, optimizer, batch size, accumulation, learning rate, epochs, and seeds. The latter reduces the headline recipe confound, although it remains a validation development comparison because both arms select their checkpoints on validation. The implementation records teacher outputs, manifests, retrieval signatures, and prompt-format hashes. Gold evidence is used only for retrieval diagnostics, never injected into P1 validation or test prompts.

4.2 Asymmetric NEI deferral (AND)

The policy uses the direct expert as an anchor:

$$\hat{y} = \begin{cases} \text{NEI}, & \arg \max_y p_G(y) = \text{NEI} \wedge p_G(\text{NEI}) \geq \tau, \\ \arg \max_y p_D(y), & \text{otherwise.} \end{cases}\tag{2}$$

We use $\tau = 0.49$, selected on validation and frozen thereafter for the registered $K = 5$ router. This rule is intentionally asymmetric: the rationale-trained expert is not permitted to overwrite a direct Supported or Refuted decision with another decisive class. It can only defer to NEI when its complementary NEI posterior is high. AND is therefore distinct from posterior averaging, symmetric confidence routing, and an unconstrained expert switch. $p_G(\text{NEI})$ is a learned class posterior, not a Bayesian estimate of parameter uncertainty or a directly validated sufficiency probability.

4.3 Disagreement-aware distillation

CURE-Student is a secondary single-model variant. Let q_D and q_G be frozen teacher distributions generated without the held-out fold. For agreement cases, the distillation target is their average; for disagreement cases, a train-only cross-fitted outcome identifies which teacher target to transfer. The confirmed variant minimizes

$$\mathcal{L}_{\text{DKD}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{KD}} T^2 \text{KL}(q \parallel p_S).\tag{3}$$

Here q is the selected teacher target and $T = 2$ is the distillation temperature. A counterfactual sufficiency loss was tested during the Fold-0 mechanism screen but was not retained in the confirmed disagreement-KD variant. CURE-Student is reported as a compute-efficient secondary extension, not as a replacement for AND.

4.4 Competitive ensemble variant

The CURE-Ensemble test system is a complementary, validation-selected ensemble: five frozen

Split/protocol	Claims	Labels used for selection	Role
Train	11,631	train only	optimization
Validation	1,456	validation	frozen policy selection
Official test	2,442	no	primary benchmark
Strict test	2,434	no	duplicate-robustness

Table 1: Protocol sizes. Official and strict test results are not pooled.

direct seeds plus the three best rationale-trained seeds ranked on validation. The composition $\{100, 87, 42\}$ is fixed before the official test is evaluated, and its prediction is the arithmetic mean of member verdict probabilities. We also report an unpruned ten-member ensemble (all five direct plus all five rationale-trained seeds) as a no-subset-selection control. This system is a competitive benchmark variant, not an implementation of AND: it answers whether heterogeneous expert predictions are complementary, whereas AND answers whether a constrained NEI deferral is useful. We do not have an eight-seed direct-only ensemble control; separating expert heterogeneity from ensemble size would require three additional independently trained direct seeds.

5 Experimental Setup

5.1 Data and preprocessing

MOCHEG contains 15,601 English claims from PolitiFact and Snopes with textual, visual, and social evidence (Yao et al., 2023). Our P1 experiments use a fixed MOCHEG *text* corpus. Qwen3-Embedding-4B retrieves a top-50 dense candidate pool, and Qwen3-Reranker-4B then scores claim-passage utility with an instruction requiring entity, temporal, quantity, location, and negation compatibility. The registered router consumes the top five reranked passages ($K = 5$), truncated to 2,200 evidence characters. No gold evidence is injected at validation or test. The open-web C-series snapshot is intentionally excluded from P1 effectiveness tables because it is a different deployment regime.

5.2 Models and metrics

All verifiers use Qwen3-4B-Instruct-2507 with LoRA adapters. The direct ensemble uses seeds $\{13, 21, 42, 87, 100\}$; the rationale-trained ensemble contains the validation-selected seeds $\{100, 87, 42\}$. The representative direct-baseline runs use LoRA rank 16, LoRA scaling 32, dropout

0.05, learning rate 10^{-4} , AdamW weight decay 0.01, warm-up ratio 0.05, three epochs, per-device batch size 1, and gradient accumulation 16. The headline rationale-trained runs use the same LoRA rank/scaling/dropout but batch size 2, accumulation 4, and learning rate 2×10^{-4} . The fresh headline-recipe direct-only control uses the latter recipe with the same three seeds and disables only the rationale objective. Exact run configurations are retained in JSON summaries.

We report accuracy, class F1, Macro-F1, expected calibration error (ECE) when available, paired percentile bootstrap intervals, and exact McNemar tests. Bootstrap samples are drawn at the claim level with 10,000 iterations. We call $P(\Delta > 0)$ a bootstrap probability of superiority, not a frequentist p -value. All test table deltas compare predictions with matched claim IDs.

5.3 Baselines and comparison discipline

Baselines include direct-only, rationale-trained-only, equal posterior averaging, symmetric confidence routing, no-asymmetric-NEI routing, disagreement-only routing, and max-confidence routing. Oracle-routing diagnostics are omitted because they do not constitute an implementable baseline. Literature scores are reported in a separate protocol-aware table. MOCHEG, HGTMFC, AMuFC, MEVER, M-RAV, and open-web systems do not necessarily share evidence, split, modality, or metric definitions; therefore a row is not treated as a direct leaderboard comparison unless its protocol is explicitly aligned.

5.4 Promotion gates and split discipline

We report a consistent directional-bootstrap reference threshold of .95, together with percentile intervals and exact McNemar tests; it is a reporting criterion rather than a claim of preregistration in this anonymized manuscript. The later duplicate-family-safe development studies use train-only folds, fixed training epochs, compute-matched controls, and fresh confirmation folds. A screen-only gain therefore cannot become an official-test claim. CURE-Ensemble is selected on validation and applied once to test. The post-hoc $\tau = .60$ sensitivity result is reported separately and never used as a frozen-test result. The fixed AND condition uses $K = 5$ and $\tau = .49$; a later K sweep is descriptive rather than a retrospective change to this condition.

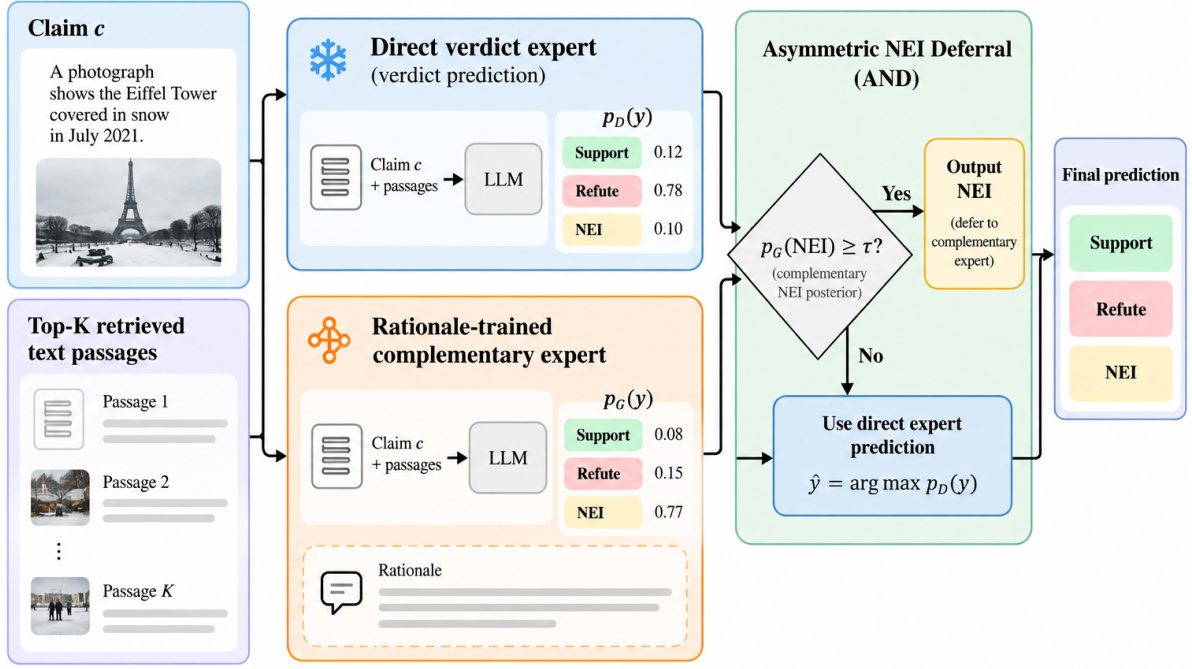


Figure 1: Overview of CURE. The claim and retrieved evidence are processed by (i) a direct verdict expert and (ii) a rationale-trained complementary expert. AND may only defer a decision to NEI; otherwise it preserves the direct prediction. The lower CURE-Student branch is a secondary distillation extension, not a required component of AND.

6 Main Results

6.1 Mechanism-focused evidence

We report only the development evidence needed to assess the proposed mechanism in the main paper. Earlier interventions are retained as a compact appendix ledger. They establish four negative findings: evidence compression did not reliably replace document context; symmetric auxiliary objectives and PCGrad did not generalize; calibration and unconstrained routing alternatives were unstable; and counterfactual curricula failed matched-control confirmation. These failures motivate retaining a competent direct verifier and treating the rationale-trained branch as a constrained NEI signal rather than a universally superior classifier.

6.2 Frozen validation and ablations

Reviewer controls. We additionally compare AND with a direct-only self-deferral baseline. The latter outputs NEI when $p_D(\text{NEI}) \geq .23$; its threshold is selected on official validation and frozen before either test track is read. Table 4 shows that AND exceeds this simpler uncertainty

Development question	Confirmed observation
Evidence compression	Atomic evidence loses discourse context; a rationale-distilled selector does not exceed retrieval Top-3.
Symmetric multi-task learning	B6 auxiliary transfer and PCGrad fail their fresh-fold or bootstrap promotion gates.
Calibration and routing	Logit bias, cross-fit calibration, and unconstrained routers produce small, non-robust gains.
Counterfactual supervision	A +.01654 Fold-0 screen gain becomes -.00104 versus matched control in confirmation.

Table 2: Four negative-result families motivating the constrained design. Detailed phase-level results are in Appendix A.

policy by .500 MF1 points on raw P1 and .523 points on strict P1. The directional advantage is replicated on both tracks, but its paired bootstrap probabilities (.7805/.7920) and intervals are not sufficient for a conventional significance claim. Thus, this control rules out a direct NEI threshold as a complete account of the observed AND point estimate, while leaving the magnitude of the incremental benefit uncertain.

Variant	Acc.	Macro-F1	F1-S	F1-R	F1-NEI
A0 Direct-only	.700549	.692045	.720348	.769904	.585882
A1 Rationale-trained-only	.706731	.701191	.704519	.780834	.618221
A2 Equal posterior average	.699863	.692773	.707071	.772608	.598639
A3 Symmetric confidence switch	.698489	.691095	.706145	.771930	.595211
A4 Unconstrained confident switch	.710165	.704436	.707904	.781915	.623491
CURE (AND)	.714973	.711205	.703016	.793008	.637591

Table 3: Frozen validation component ablations (validation is used only to evaluate fixed policies). “Unconstrained confident switch” selects the rationale-trained output of any class when its maximum probability is at least τ . CURE applies the class-constrained AND rule at $\tau = .49$. Degenerate disagreement-only and max-confidence variants are omitted from the main table and reported in the accompanying ablation artifact. F1-S/R denote Supported/Refuted.

Track	Comparison	Base MF1	AND MF1	Δ	$P(\Delta > 0)$
Validation	AND vs. rationale-trained-only	.70119	.71121	+0.01001	.9470
Validation	AND vs. unconstrained switch	.70444	.71121	+0.00677	.8952
Raw P1	AND vs. direct	.54493	.55470	+0.00977	.9529
Raw P1	AND vs. self-deferral	.54970	.55470	+0.00500	.7805
Strict P1	AND vs. direct	.54490	.55433	+0.00943	.9477
Strict P1	AND vs. self-deferral	.54911	.55433	+0.00523	.7920

Table 4: Frozen-policy reviewer controls. Self-deferral uses only the direct expert with validation-selected $\tau_D = .23$; AND uses $K = 5, \tau = .49$. Raw/strict rows use no test labels for policy selection. The bootstrap probability is directional evidence, not a substitute for a confidence interval.

Broader prior-work inventory. Table 5 lists twelve system rows: eight prior-work rows and four rows from this work. It is deliberately *not* a single leaderboard: a reported F-score is preserved as F-score rather than silently renamed Macro-F1; filtered tests, gold evidence, or alternate preprocessing are marked. Only rows labelled “aligned raw P1” support an apples-to-apples point-estimate comparison.

AND has the highest frozen-validation point estimate in Table 3. Relative to direct-only it gains 1.916 Macro-F1 points and 1.442 accuracy points. Replacing the class-constrained rule by an unconstrained confident switch reduces Macro-F1 by 0.677 points, whereas equal averaging loses 1.843 points. The rationale-trained-only branch is useful (+0.915 over A0) but remains 1.001 points below AND. AND’s validation comparisons against rationale-trained-only ($P = .9470$) and unconstrained switching ($P = .8952$) do not independently clear the reported .95 directional-bootstrap reference threshold; they are point-estimate evidence, not decisive mechanism confirmation. The raw and strict P1 results retain the direction of the direct comparison, while their uncertainty intervals require a cautious interpretation.

6.3 Official test results and leakage control

The frozen CURE-Ensemble reaches Macro-F1 0.55507 and accuracy 0.57535 on raw P1 official test, compared with 0.54531 and 0.56798 for the previously materialized direct five-seed ensemble. The paired bootstrap probability of improvement is 0.9839 with CI [+0.00097, +0.01850]. Its score is higher than AMuFC’s reported raw-P1 point estimate (Macro-F1 .5400; accuracy .5460), but this is not a paired comparison and the two systems differ in retrieval, backbone, modality use, and member count. We therefore report it as a protocol-aligned point estimate rather than a claim of statistical superiority or state of the art. CURE-Student disagreement distillation reaches Macro-F1 0.54940 for a post-hoc reported single seed (87); it is an exploratory efficiency diagnostic, not a multi-seed or locked-test effectiveness comparison.

AND has canonical test artifacts on both tracks. On raw P1, AND obtains .55470 Macro-F1 and .57125 accuracy, compared with its independently inferred direct anchor’s .54493 and .56798: $\Delta\text{MF1} = +.00977$, with 86 helpful and 78 harmful changes and paired bootstrap $P(\Delta > 0) = .9529$. On strict P1, it obtains .55433 versus .54490 ($\Delta\text{MF1} = +.00943$; 84/77 helpful/harmful; $P(\Delta > 0) = .9477$). The direct-only self-deferral control is weaker on both tracks (.54970 raw and .54911 strict); AND improves on it by +.00500/+0.00523 MF1, respectively. The associated probabilities (.7805/.7920) and all test-track percentile intervals include zero, and McNemar tests are non-significant. We therefore treat the replicated direction as modest directional test evidence, not as statistically resolved confirmation.

Artifact alignment. The .00039 raw-MF1 difference between the legacy direct ensemble (.54531) and the independently inferred AND anchor (.54493) arises because the latter is a new canonical raw-manifest inference artifact built to align exactly with the AND members. It is not used in a cross-artifact paired comparison: every AND comparison uses only predictions materialized together on the same manifest. Because raw and strict ranked evidence differ for some shared IDs, each track is inferred independently from the same frozen checkpoints; no intersection or test-time tuning is used.

Validation–test gap. The larger validation–test difference is not interpreted as a model improve-

System	Venue/year	Evidence and evaluation	Models	Acc.	Reported F1	Comparability
MOCHEG (Yao et al., 2023)	SIGIR 2023	system text+image	–	–	F-score .4406	original benchmark; score label retained
LVL4M4FV (Tahmasebi et al., 2024)	arXiv 2024	retrieved text+image	–	.4510	micro-F1 .4510	macro not source-reported
HGTMFC (Pang et al., 2025)	AAAI 2025	retrieved text+image, raw test	–	.4861	Macro-F1 .4678	aligned split; different model/retrieval
MetaSumPerceiver (Chen et al., 2024b)	ACL 2024	system text+image	–	–	F-score .4860	source does not label it
MEVER (Zhang et al., 2026)	arXiv 2026	retrieved multimodal	–	.483±.021	MF1 .497±.012	Macro-F1 custom consistent pre-processing
M-RAV (Qwen2.5-32B) (Luu et al., 2027)	IPM, forthcoming	system evidence, filtered $n = 2001$	–	.5002	MF1 .5014	filtered test; not aligned
Entailed Opinion (Kumar et al., 2025)	arXiv 2025	domain-generalization setting	–	–	MF1 .5700	not P1 system retrieval
AMuFC (Jung et al., 2026)	HCAI/CIKM 2025; arXiv 2026	retrieved text+image, raw test	–	.5460	MF1 .5400	aligned raw P1 challenger
CURE Direct ensemble	this work	fixed corpus, raw P1 $n = 2442$	5	.5680	MF1 .5453	aligned raw P1
CURE rationale seed 100	this work	fixed corpus, raw P1 $n = 2442$	1	.5696	MF1 .5513	aligned raw P1, one seed
CURE AND	this work	fixed corpus, frozen $K = 5, \tau = .49$, raw P1	8	.5713	MF1 .5547	aligned raw P1, canonical router
CURE-Ensemble	this work	validation-selected, raw P1 $n = 2442$	8	.5754	MF1 .5551	aligned raw P1, frozen-test variant

Table 5: Protocol-aware comparison with prior work. “Models” is the number of sequential member forward passes when reported; a dash means not source-reported. “Aligned” means raw MOCHEG P1 test with system-retrieved evidence; it does not assert identical retrievers or model capacity. Only aligned rows may be used for a direct point-estimate comparison.

Evidence policy	Acc.	Rationale MF1	AND MF1
$K = 1$	.707418	.660969	.704452
$K = 3$	.718407	.695423	.715270
$K = 5$	.714973	.701191	.711205
Direct-only	.700549	–	.692045

Table 6: Evidence-count sensitivity on validation. The fixed AND condition uses $K = 5$, inherited from the direct retrieval protocol. The later $K = 3$ sweep has the highest validation score but is not substituted for the fixed condition or used for an official-test claim.

τ	Validation MF1	Status
.40	.70774	sensitivity
.45	.70932	sensitivity
.49 (frozen)	.71121	registered AND
.50	.71109	sensitivity
.55	.71089	sensitivity
.60	.71012	sensitivity only
.65	.70755	sensitivity
.70	.70420	sensitivity

Table 7: Validation threshold sensitivity for AND. $\tau = .49$ is frozen for the registered condition. No raw-test threshold sweep is used for parameter selection.

ment. Validation was used for seed and threshold development and has a distinct claim composition. The contraction from +.01916 MF1 on validation to +.00977/+0.00943 on raw/strict test is consistent with development-selection sensitivity, but this study cannot apportion it among distribution shift, optimization, and selection.

Routing versus averaging. Raw CURE-Ensemble (.55507) is marginally above raw AND

CURE system	Macro-F1	Accuracy	Raw-P1 status
Direct five-seed ensemble	.5453	.5680	legacy frozen baseline artifact
CURE rationale-trained seed 42	.5338	.5524	one seed
CURE rationale-trained seed 87	.5394	.5606	one seed
CURE rationale-trained seed 100	.5513	.5696	best single rationale-trained seed
CURE-Ensemble full 10-member	.5512	.5717	unpruned control
CURE-Ensemble selected 8-member	.5551	.5754	frozen-test benchmark variant
CURE AND (5+3 experts)	.5547	.5713	canonical frozen router
CURE-Student DKD seed 87	.5494	.5676	raw P1, one seed

Table 8: Raw official P1 test ($n = 2,442$) for frozen CURE system variants. CURE-Ensemble and AND use distinct, validation-frozen compositions/policies and should not be read as interchangeable ablations. AND uses fixed $K = 5, \tau = .49$ with independently inferred raw-P1 predictions. The legacy direct artifact (.54531 MF1) is not the paired AND anchor; Section 6.3 uses a jointly materialized canonical direct anchor (.54493 MF1).

(.55470), although A2 averaging is unhelpful on validation. This difference does not establish that either decision rule dominates: the systems have distinct validation-frozen compositions and objectives, and neither is an ablation of the other. Both reports record manifest, retrieval, input-prediction, and materialized-output hashes.

Evidence policy	Acc.	MF1	F1-S	F1-NEI
Original Top-5 anchor	.6664	.6435	.6026	.5067
Retrieval Top-3	.6720	.6482	.5964	.5187
Generic CE selector	.6496	.6374	.5674	.5253
Rationale-distilled selector	.6651	.6460	.6098	.5103

Table 9: Sentence-selection train-only Fold-0 screen ($n = 2,326$). The rationale-distilled selector is better than generic CE but does not exceed retrieval Top-3.

7 Additional Ablations and Negative Results

7.1 Sentence-selection branch

A sentence-selection ablation distills teacher key-passage attribution into a MiniLM cross-encoder that adaptively selects one to three sentences. It is an evidence-selection ablation, not the dual-expert router. On Fold 0, rationale-distilled selection improves a generic cross-encoder by $+0.00865$ MF1 but remains -0.00214 behind the retrieval Top-3 control. It was therefore not expanded to more seeds or test evaluation (Table 9). This result prevents us from attributing the rationale-trained-expert effect to sentence compression alone; teacher capacity is instead isolated by the matched full-train ablation below.

7.2 Why earlier constraint approaches were not promoted

Earlier interventions supplied useful diagnostics but did not establish a stable main method. A source-aware auxiliary ensemble reached .7080 MF1 on validation, yet its bootstrap probability against the compute-matched direct control was .9354, below the reported .95 reference threshold. Calibration and curriculum variants exhibited strong Fold-0 gains but degraded under fresh-fold confirmation. A counterfactual verdict curriculum improved Fold 0 by $+0.01654$ MF1 versus its anchor, but aggregated confirmation was -0.00104 versus the matched control. These negative outcomes motivate the CURE design choice: preserve a competent direct decision model and use a rationale-trained complementary signal through a constrained interface rather than continuously modifying the anchor’s decision boundary.

7.3 Matched-recipe explanation supervision control

This is a distinct control from the headline-recipe contrast in Table 10. We ran a five-fold train-only OOF experiment in which verdict-only and

Ensemble	Acc.	MF1	F1-S	F1-R	F1-NEI
Headline-recipe direct-only	.69437	.68712	.69874	.77003	.59259
Rationale-trained	.70673	.70119	.70452	.78083	.61822

Table 10: Headline-recipe direct-only control on official validation ($n = 1,456$). Rationale-trained minus direct-only: $\Delta\text{MF1}=+.01407$, helpful/harmful=88/70, bootstrap 95% CI $[-.00369, +.03180]$, and $P(\Delta > 0) = .9396$. Both arms use validation for checkpoint selection; no test result is selected or evaluated here.

explanation-supervised models use one common training script and optimizer schedule. The explanation model obtains .65757 aggregate MF1 versus .65762 for the matched verdict-only control ($\Delta = -.00006$; bootstrap $P(\Delta > 0) = .4940$). Applying the frozen asymmetric router to this matched explanation expert yields .65898, only $+0.00135$ over the control ($P(\Delta > 0) = .7273$). These OOF values are neither validation nor test results and cannot be directly compared with the .71121 validation policy score. They show that structured rationale loss alone does *not* improve average verdict accuracy under that common recipe.

7.4 Headline-recipe direct-only control

To reduce the optimization-recipe confound, we train a new verdict-only ensemble with the exact headline rationale-trained recipe: the same data, no gold injection, batch size 2, gradient accumulation 4, learning rate 2×10^{-4} , three epochs, top- $K = 5$, and seeds $\{42, 87, 100\}$. The only intended difference is $\lambda_r = 0$ versus .25. On validation, the rationale-trained ensemble is higher by $+0.01407$ MF1 and $+0.01236$ accuracy, with its largest class gain on NEI ($+0.02563$; Table 10). This directional contrast reduces the explanation-versus-recipe ambiguity, but does not settle it statistically: the paired 95% interval includes zero, $P(\Delta > 0) = .9396$, and both arms selected their best checkpoint on the same official validation split.

We then freeze these same three seeds and independently infer raw and strict P1 predictions. The rationale-trained ensemble retains a positive point-estimate difference on raw ($+0.00512$ MF1) and strict ($+0.00685$ MF1) test (Table 11); the directions therefore replicate across both test manifests. However, their intervals contain zero ($P(\Delta > 0) = .7694/.8270$). The full-train/test control is consequently evidence compatible with a rationale contribution, rather than statistically resolved causal confirmation.

Track	Direct MF1	Rationale MF1	Δ	Helpful/Harmful	95% CI	$P(\Delta > 0)$
Raw P1	.53474	.53986	+0.0512	119/109	[-.00856, .01876]	.7694
Strict P1	.53222	.53906	+0.0685	120/106	[-.00705, .02049]	.8270

Table 11: Frozen full-train B20 confirmation. The same three validation-frozen seeds are independently inferred on raw and strict P1; no test label selects a checkpoint, seed, or policy. Positive test point estimates replicate, but neither paired interval excludes zero.

Track	Teacher	Acc.	MF1	F1-S	F1-R	F1-NEI	$P(7B > 3B)$
Raw P1	Qwen2.5-3B	.56839	.54829	.54705	.65437	.44346	—
Raw P1	Qwen2.5-7B	.55979	.53986	.52416	.65332	.44209	.1166
Strict P1	Qwen2.5-3B	.56738	.54660	.54633	.65466	.43881	—
Strict P1	Qwen2.5-7B	.55957	.53906	.52424	.65421	.43875	.1365

Table 12: Matched teacher-capacity ablation with identical student, data, retrieval, recipe, and seeds. For 7B minus 3B, raw/strict Δ MF1 are $-.00844 / -.00753$; 95% CIs are $[-.02223, .00562] / [-.02147, .00626]$. Test labels select neither teacher, checkpoint, seed, nor policy.

7.5 Matched teacher-capacity ablation

The capacity alternative is tested directly rather than inferred from the headline result. We regenerate only the training rationales with Qwen2.5-3B-Instruct, retrain the same Qwen3-4B students with the same explanation weight, data, retrieval, prompts, seeds {42, 87, 100}, optimizer recipe, and validation checkpoint protocol, and independently infer both P1 tracks. Thus, the ablation changes teacher capacity but does not add data, gold evidence, student capacity, or test-time tuning.

Table 12 does not support the hypothesis that the 7B teacher’s larger capacity explains the rationale-trained branch: the 3B teacher has higher point estimates on raw (+.00844 MF1 relative to 7B) and strict (+.00753). This direction is consistent across the two independently inferred manifests. It is nevertheless not a statistically resolved claim that 3B is superior: the paired 95% intervals for 7B minus 3B include zero and $P(\Delta > 0) = .1166 / .1365$. The defensible conclusion is narrower: within this matched teacher family and protocol, scaling the teacher from 3B to 7B is not evidence for the observed rationale-branch behavior. A plausible explanation is that a smaller teacher produces less noisy or more concise targets, but this remains a hypothesis rather than an identified mechanism.

8 Efficiency and Robustness Analysis

8.1 Inference efficiency

On an RTX 5090 with 256 common validation claims, batch size 4, and two warm-up batches,

System	Passes	ms/claim	claims/s	Peak GiB
Direct ensemble	5	320.72	3.118	14.74
Rationale-trained ensemble	3	193.17	5.177	14.74
AND dual ensemble	8	513.88	1.946	14.74
CURE-Student	1	64.47	15.511	14.74

Table 13: Efficiency benchmark: RTX 5090, 256 claims, batch size 4, two warm-up batches. Model-loading time is excluded.

Group	n	Δ MF1	Helpful/Harmful
Overall	1,456	+0.01916	58 / 37
PolitiFact	431	-.00249	0 / 6
Snopes	1,025	+0.01994	58 / 31
Gold NEI	478	+0.03239	58 / 0
Gold Refuted	487	-.00343	0 / 9
Gold Supported	491	-.01404	0 / 28
Gold rank 1	1,233	+0.02772	54 / 25
Gold rank 2-5	131	-.02607	3 / 7
Gold not retrieved	59	-.03260	1 / 4

Table 14: Selected frozen-validation subgroup audit: AND minus direct. Gold rank is a diagnostic label unavailable at inference, not a routing input.

direct, rationale-trained, AND, and the one-model student require 320.72, 193.17, 513.88, and 64.47 ms per claim, respectively (Table 13). AND is therefore $1.60\times$ slower than its five-pass direct anchor for only a $+.00500 / +.00523$ unresolved advantage over direct self-deferral on raw/strict test. The student uses one pass instead of the router’s eight sequential passes. All ensemble timings are sequential; the repeated 14.74 GiB figure is the maximum allocated memory of one loaded member, not a comparison of total ensemble memory. Subset effectiveness scores produced by this benchmark are diagnostic only.

8.2 Subgroup behavior

On the frozen validation audit, AND improves Macro-F1 by 1.916 points, with 58 helpful and 37 harmful changes. Table 14 separates where this gain occurs. It is largest for NEI (+3.239), rank-one evidence (+2.772), evidence hits (+2.280), and Snopes (+1.994). Retrieval misses decrease by 3.223 points and Supported claims decrease by 1.404 points. These are post-hoc descriptive groups, based partly on diagnostics unavailable at inference. They do not establish that the rationale-trained posterior measures corpus-level insufficiency, particularly under retrieval failure.

8.3 CURE-Student distillation

Five-fold train-only OOF confirmation gives the student control Macro-F1 0.660788 and disagreement-KD Macro-F1 0.667607, a

Fold-0 variant	Acc.	MF1	F1-S	F1-NEI
Matched control	.6690	.6570	.6013	.5368
Naive ensemble KD	.6707	.6513	.6106	.5152
Counterfactual only	.6758	.6577	.6063	.5332
Full KD + disagree + CF	.6776	.6584	.6022	.5359
Disagreement KD	.6836	.6687	.6276	.5472

Table 15: CURE-Student train-only Fold-0 mechanism screen ($n = 2,326$). The five-fold confirmation uses disagreement KD only; counterfactual terms are reported solely as screen ablations.

+0.006819 difference. Four of five folds improve and paired bootstrap $P(\Delta > 0) = 0.9517$. Table 15 shows why the disagreement target matters: naive ensemble KD is worse than the matched control, while counterfactual loss alone is nearly neutral and the full objective reduces the screen gain. This supports CURE-Student as a practical compression extension, but its margin is close to the reported bootstrap reference threshold. The reported seed-87 test number is post-hoc descriptive among the evaluated student seeds, not a locked-test model-selection claim.

9 Error Analysis

The dominant helpful transition is direct Refuted/Supported to rationale-trained NEI on gold NEI claims. Harmful transitions are direct Supported to NEI (28 cases) and direct Refuted to NEI (9 cases). This explains why AND improves NEI while reducing decisive-class F1. The failure atlas also shows that routing under missing or rank-2–5 evidence is less reliable. A future system should estimate retrieval reliability before allowing deferral, but using gold rank or QREL status at inference would be invalid.

10 Discussion

The results support a narrow interpretation of the contribution. Rationale-trained and direct experts form a complementary pair, but explanation supervision alone is not consistently beneficial: a matched explanation-loss ablation produced approximately zero aggregate gain. The headline-recipe control is directionally favorable to rationale training, yet its validation and test intervals include zero. The matched B21 result further weakens a capacity-based explanation: replacing the 7B rationale teacher with its 3B counterpart raises the raw/strict point estimates by .844/.753 MF1, but neither comparison resolves statistically. Accordingly, the evidence does not identify a causal mech-

anism within rationale supervision, and it does not support the claim that a larger teacher is responsible for the branch’s behavior.

Moreover, a decisive-evidence coverage diagnostic finds that the direct posterior has stronger ranking and calibration than the rationale-trained posterior (AUROC .6486 vs. .6155; Brier .1122 vs. .1406); this diagnostic uses gold labels only after prediction and is never a router input. AND also exceeds a frozen direct self-deferral baseline on both test tracks, although that incremental comparison is not statistically resolved. The useful object is therefore the asymmetric interface between complementary class posteriors, not a demonstrated sufficiency estimator. This interface also exposes a compute-quality trade-off: AND uses eight sequential expert passes, whereas CURE-Student compresses complementary teacher behavior into one student. A stronger external definition of evidence sufficiency remains necessary before claiming that either posterior is calibrated for that construct.

11 Limitations and Ethical Considerations

MOCHEG is a single benchmark with two principal sources and potentially duplicated claims; we therefore report raw and strict tracks. The fixed-corpus protocol does not measure live-web robustness or establish multimodal effectiveness. The rationale-trained expert inherits teacher errors and pseudo-label noise; moreover, teacher rationales are conditioned on the gold *training* verdict, which is appropriate for training but prevents the rationale channel from being interpreted as independent supervision. Although B21 controls the 3B-versus-7B capacity comparison, it evaluates only two capacities from one teacher family and cannot identify why their target distributions differ. The headline-recipe direct control reduces causal ambiguity and preserves a positive point estimate on both test tracks, but its validation and test intervals include zero. Subgroup analyses are descriptive and do not justify post-hoc policy tuning. Although AND now has independently inferred raw and strict test artifacts, its 95% paired-bootstrap intervals include zero and strict-track directional evidence is .9477; stronger multi-dataset confirmation is needed before broad generalization claims. External-dataset evaluation and a broader teacher-family study remain future work. Automated fact checking can amplify source and retrieval bias; pre-

dictions should be treated as decision support, not authoritative adjudication.

12 Conclusion

We presented CURE, a dual-expert constrained routing framework for retrieved-evidence fact verification. Its AND rule improves frozen-validation Macro-F1 by 1.92 points and has raw/strict P1 test point-estimate gains of .977/.943 points under a fixed $K = 5$, $\tau = .49$ policy. A same-recipe direct-only control retains positive raw/strict rationale-trained differences of .512/.685 MF1 points, but their intervals include zero. A matched 3B-versus-7B teacher-capacity ablation does not support simple teacher scaling as the explanation: 3B has higher raw/strict point estimates, while the paired intervals remain unresolved. Thus the routing, recipe-control, and capacity evidence should be read as directional rather than statistically resolved causal proof. CURE-Ensemble provides a separate validation-selected benchmark variant, while CURE-Student demonstrates a one-model compression extension. The next research steps are external-dataset evaluation, broader teacher-family controls, and sharper retrieval-reliability gating, rather than post-hoc policy tuning on P1.

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A Detailed Development Ledger

Table 16 expands the main trajectory with the numerical evidence used to close or advance each hypothesis. These are reported for reproducibility and to prevent failed development screens from being misrepresented as official-test improvements.

B Five-Fold CURE-Student Confirmation

C Raw versus Strict Test Reporting

The raw official test ($n = 2,442$) is the primary literature-comparison track. The strict test ($n = 2,434$) removes eight verbatim cross-split duplicates and is a robustness track only. Table 18 does not merge the two populations. Canonical AND uses independently inferred predictions on each retrieval manifest because their ranked evidence IDs differ for some shared claims; every prediction file is checked against its full manifest rather than evaluated on an intersection.

D Reproducibility Checklist

The anonymized artifact package records manifests, retrieval signatures, prediction files, training summaries, seeds, and protocol flags. It includes training, component-ablation, subgroup, distillation, efficiency, and canonical-evaluation entry points. Before camera-ready release, we will add exact GPU driver versions, package-lock information, author metadata, and a public archival link.

ID	Intervention	Quantitative observation	Interpretation
SV	flat versus hierarchical sufficiency verification	Flat MF1=.6371; SV=.6398 (+.0027) on Fold 0; five-fold confirmation +.00060 $\pm$.00745	Hierarchy had small local benefit but no stable promotion.
DRO	source/evidence DRO	qrel-absent MF1 +.1004; overall MF1 -.0021	Robustness improvement cannot be bought with aggregate regression.
Atomic	atomic sentence verifier	sentence retrieval Recall@8=.8661; verdict MF1 -.0270 vs article evidence	Retrieval coverage alone is insufficient; discourse matters.
Packets	context packets	best interpolation +.01014 on one validation split; five-seed packet gain +.00387 $\pm$.00862	Complementarity exists but is not a reliable replacement.
Multi-task	sufficiency/polarity multi-task	auxiliary vs matched control +.00886 $\pm$.01081; bootstrap .9354	Positive transfer signal, below the reported reference threshold.
PCGrad	PCGrad	conflict rate about 45–46% in epoch 1; ensemble -.0105 vs control	Gradient conflict was real but projection did not improve generalization.
Router	selective expert router	MF1 +.00460; bootstrap .9294	Helpful precision, insufficient gain and confidence.
Bias	class-logit bias	MF1 +.00216; bootstrap .7604	Prior shift does not explain the main error pattern.
Ensemble	fixed seed ensemble	Fold-0 +.01348; fresh-fold mean +.00398	A screen peak is not confirmation.
Calibration	cross-fitted calibrator	aggregate MF1=.67054; +.00345 vs unweighted ensemble	Bootstrap/source gates fail.
Provenance	provenance calibrator	-.00176 vs unweighted ensemble	Provenance-conditioned calibration harms aggregate accuracy.
Constraints	joint constraints from base	+.00538 vs compute control but -.01376 vs anchor	Constraint tasks help relative to compute but induce anchor drift.
Curriculum	curriculum/blend	Fold-0 blend +.00510; five-fold mean +.00090	Direct curriculum is unstable.
NEI escape	NEI escape analysis	MF1 +.00209; accuracy +.01126	It helps supported/refuted escape but damages NEI under qrel absence.
Value gate	value-of-information gate	MF1 +.00380; AUROC=.5660	Observable utility is too weak for reliable routing.
Counterfactual	counterfactual omission	Fold-0 +.01654 vs anchor; confirmation -.00104 vs control	The strongest screen still fails matched-control confirmation.
Atlas	failure atlas	candidate/control .66517/.66621; help/harm 550/561	No stable subgroup justified further tuning on inspected folds.

Table 16: Earlier negative and diagnostic experiments. Each value remains in its native protocol; no row is a P1 official-test claim.

Fold	Control MF1	DKD MF1	Δ
0	.657035	.668656	+.011621
1	.659998	.656767	-.003231
2	.662265	.667143	+.004878
3	.647130	.660188	+.013058
4	.674931	.681645	+.006714
Aggregate	.660788	.667607	+.006819

Table 17: CURE-Student train-only out-of-fold confirmation. Four of five folds improve; the aggregate paired bootstrap probability is .9517.

System	Track	Acc.	MF1
Direct five-seed	raw official	.56798	.54531
Direct five-seed	strict	.56902	.54581
CURE-Ensemble	raw official	.57535	.55507
CURE-Ensemble	strict	.57477	.55383
CURE-Student seed 87	raw official	.56757	.54940
CURE-Student seed 87	strict	.56615	.54733
AND router	raw official canonical	.57125	.55470
AND router	strict canonical	.57149	.55433

Table 18: Track-separated test results. AND uses fixed $K = 5$, $\tau = .49$ and independently inferred raw/strict predictions; it is not an intersection evaluation.